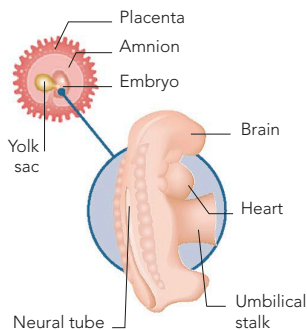


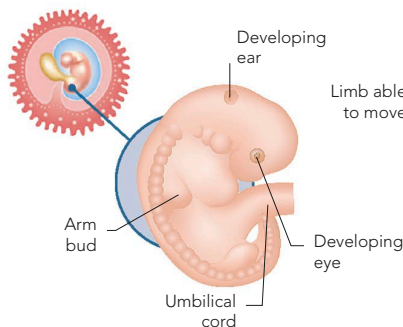
## GROWING EMBRYO

In general, development is head-down: the brain and head take shape early, then the body, the arms, and lastly the legs. Eight weeks after fertilization, all major organs and body parts have formed. From this time on, the baby is known as a fetus.



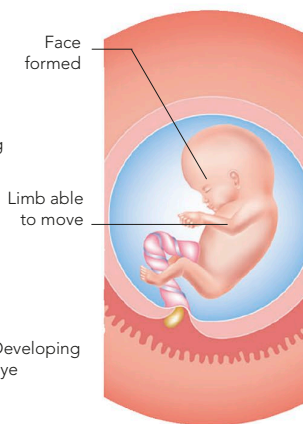
### THREE WEEKS

The embryo is 2–3mm ( $\frac{1}{50}$ – $\frac{1}{50}$  in) long. The neural tube forms. It will become the spinal cord, with a brain at one end. A tube-like heart pulsates.



### FOUR WEEKS

The embryo is about 4–5mm ( $\frac{1}{5}$  in) long. A four-chambered heart beats, sending blood through simple vessels. Intestines, liver, lungs, and limb buds can be seen.

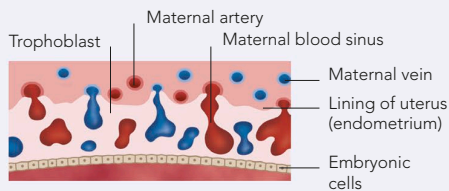


### EIGHT WEEKS

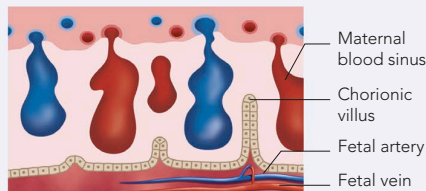
The embryo is around 25–30mm (1–1  $\frac{1}{5}$  in) long. The face, neck, fingers, and toes can be seen.

## DEVELOPMENT OF THE PLACENTA

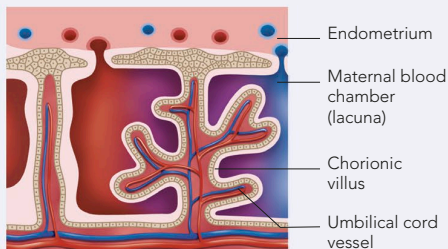
The placenta derives from the trophoblast – the outer layer of the blastocyst (the mass of cells that results from the fusion of egg and sperm). It begins to form soon after the fertilized egg implants in the uterine lining (see opposite), and becomes almost fully developed by the fifth month of pregnancy.



**1** Embryonic cells extend into uterine blood vessels, so that maternal blood flows into spaces (sinuses) within the trophoblast.



**2** Fingerlike projections, called chorionic villi, grow and are surrounded by maternal blood sinuses. Later, fetal blood vessels grow into the villi.



**3** The villi branch further and the maternal blood sinuses enlarge into lacunae ("lakes"), supplying the placenta with oxygen and nutrients.